The Wireless Internet: Connecting All Of Us

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In today's wireless world, everyone is connected. More and more, your phone, PDA, or pager will become a part of your communication tools to access personal information, find local information and coordinate with friends, family and colleagues. Real-time access all the time is rapidly becoming a reality displacing the concept of personal with the concept of community.

Kevin Clark oversees all aspects of NotWired's vision and mission. He is involved in the realization of the NotWired platform and its positioning in the marketplace.

The first wave of the Internet connected millions. The mobile Internet wave will connect billions in many countries not as connected as the U.S. and Europe. Billions of users in every country around the world will interact through mobile devices with others over the wireless Internet.

Instant information and communications will be the cornerstone of the next Internet revolution, and it will affect us even more than the changes brought on by the last ten years of technology advancement.

As more and more of us move our calendar and contacts from personal organizers and onto the Internet, the potential for instant access to our personal information becomes a reality. However, where is the potential to join together in groups and interact with anyone? Today's solutions are artificially limited to personal islands of information.

We have electronic options to manage scheduling and contact information including PDAs, PC software and Internet applications. Two major architectures arise from these solutions. One solution is to connect islands of information through synchronization. The other is to create a central database and access it real-time. This whitepaper will discuss these two options.

Sync or Real-Time, All The Time

There are two schools of thought about data. One school of thought believes that your different systems should always contain an exact copy of data and always be in sync. The other school of thought believes the best form of access is to your centralized copy of real-time data that is always current and correct.

For personal information where users have a choice about where they keep data and sharing is limited, syncing is a viable option. For group or shared data, syncing is not a viable option since it does not facilitate real-time updates for all parties involved with the data.

Sync, Myth Versus Reality

Synchronization is a byproduct of complexity. When too many separately designed systems try to correctly interact with each other synchronization or data replication becomes a requirement.

Sync is no magic bullet. It has been with us from the time when computer professionals first put together two different computer systems. In IT, we call it data replication. In personal computing when used to manage personal contacts, calendars and files, we call it sync (short for synchronization).

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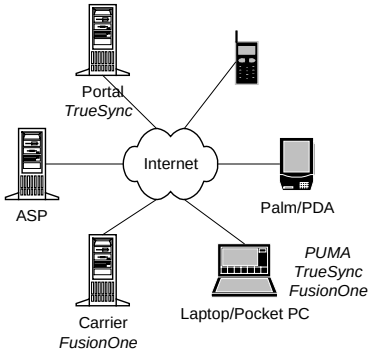
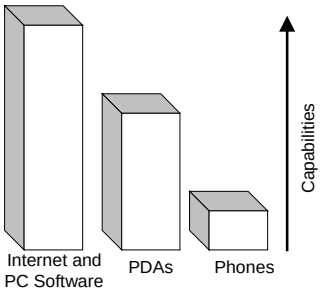
In this section we will sort out the claims about sync and put them in the context of reality.

Myth: Sync will always keep your data current.
Reality: Sync is only as good as its owner. The Palm PDA, the world's most popular PDA, ships with synchronization software. It will synchronize the Palm Desktop software with your Palm. However, most of us do not sync every day, yet our data changes every day. Why? Because synchronization takes time and it must be made a habit. Most of us forget or don't take the time to sync.



Myth: It is possible to have one sync to service all your needs.
Reality: Every device manufacturer, every software vendor and every web site chooses partners. Currently, there are a few major sync vendors in the marketplace. Most of us do not interact with just one system, therefore multiple synchronization tools will need to be installed on our systems to make synchronization even a remote reality.

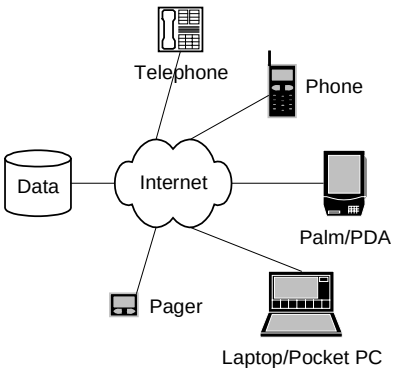
Myth: Sync is automatic and non-intrusive.
Reality: Synchronization requires a piece of software to attach to every device, software or web site. When you download some sync tool and install it on your desktop, it does not work automatically. It requires manual operation every day for every different web site, device or software that uses different sync technology.



Myth: Standardization will solve synchronization problems.
Reality: It will take time for all devices, software and web services to standardize. This is still a few years away.

Myth: Sync will handle every device and software.
Reality: Some software has features that are not supported by standards. For example, recurring events in Outlook are not the same as recurring events in ACT!. Complete sync is not possible since different software systems have different capabilities.

Real-time, All The Time



Real-time access to your data will be available from any Internet connected PC or device. By 2001, the majority of cell phones will be shipping with access to the Internet. PDA vendors like Palm, Compaq, and Handspring are rapidly delivering wireless capabilities through partners and future models include wireless capabilities. If you believe any device purchased in the next couple of years will have an antenna on it and PCs, TVs and other devices will be connected to the Internet, then real-time all the time is the best solution.

Myth: People keep separate calendars on separate systems.
Reality: Some of us do. Actually, most of us prefer to keep schedule information on one calendar.

Myth: Personal information is not shared.
Reality: People schedule with each other. They schedule business meetings, personal appointments, parties, travel, vacations and many other calendar related events with themselves and others.

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Myth: Major devices will not have wireless support.
Reality: Most major devices and PDAs will support wireless in less than two years.

Myth: I will not be able to get to my data.
Reality: Data availability is a function of connection, system uptime, backup and many other maintenance issues. Major shared systems have higher availability than the best of today's PC systems.

If you can access the latest up to the date information from any device all the time, why would you sync? Is there a better way?

Real-time, All The Time Over The Internet

The best solution is to store your information in one central location and access it from anywhere. You should be able to share information digitally with anyone and not be limited by time consuming manual sync processes or closed client/server systems.

One such common open medium to share information digitally is the Internet. Over the Internet we can create an open scheduling system to share information with colleagues, friends and family. We can securely share events, contacts, files and more, through the Internet. Listed below is a summary of the various scheduling and contact management system options and their pros and cons.

Paper-based	PIM/Sync	Client/Server Groupware	Internet Groupware
PROS			
• Status Quo	• Higher availability due to redundancy • Ability to sync work and personal systems when multiple calendars are required • Partial data updates rather than import and export	• Large install base • Familiar Windows Look & Feel	• Lower cost • Near zero administrative costs • Unlimited customization • Access through any device or any computer • No sync required • Ability to share with anyone • Ability to schedule anyone using an email or phone number • Can integrate multiple groups
CONS			
• No backup • No sharing • Bulky	• No standards • Client side software on every synced device • Slow • Error prone • Not automatic • Intrusive • Not real-time data • Manual sharing of information	• More expensive • Limited scale • Share within the organization only • Costly to maintain • Requires servers and LANs • Data not 100% available • Extra hardware and software required for voice and wireless access	• Data not 100% available when no Internet connection exists

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Real World Scenarios

Most events we arrange involve other people. Most information we have is shared with others. Group scheduling is only natural. Sharing information is natural. The Internet is the perfect open medium to share and work in groups.

In this section we will describe real scenarios encountered in our business and personal lives. We will present the sync solution and the real-time solution to the problem scenario.

Scenario 1: Scheduling a meeting with a customer
In the business world, this is something we do every day.

Sync with personal info	Real-time group info
We contact all parties involved, play phone tag or email tag and get a meeting scheduled.	Choose to create a meeting, pick people involved and send an invite. Invitees can receive the information on via email or via messages to the phone.

Scenario 2: Changing a meeting
Meetings, and other events, are dynamic by nature. Times, dates, locations change often within hours or minutes of an event. With a sync solution, invitees or members do not know about the change.

Sync with personal info	Real-time group info
Does not have the invitees concept, so the person who changes the meeting must email, message or call everyone about the change.	A significant change to the meeting will trigger an automatic notification to all involved parties according to their notification preferences.

Scenario 3: Updating a phone number on a shared phonebook
Contact information is also dynamic by nature. Family contacts, friends, business contacts and colleagues contained in our phonebooks changes every day. The more contacts you have, the more information there is to maintain. Sync and real-time have two very different requirements to maintain contact information.

Sync with personal info	Real-time group info
A change in a contact's information is sent via email, you enter it into your PDA and then initiate a sync.	A member of your group can access their profile and change information about themselves. You do nothing! The next time you access the information, it is current.

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Scenario 4: Fetching a phone number from a shared phonebook
You want to look something up in your phonebook. For business and many other groups, the phonebook can be large and constantly changing. In scenario 3, we discussed the power of real-time information. What happens when the base of information is large and changing. Personal solutions are often limited by space. For example, a phone today can hold 100 phone numbers. How often have you wanted to get a number not on your phone? And, if you are one of the few who sync your phone, how often have you not had the number you wanted?

Sync with personal info	Real-time group info
Sync your PDA, contact management system and phone. However, your PDA can hold 1,000 names, your phone holds 100 names and you contact database has 8,000 names. You better pick the right list…	Simply go to the lookup and get the phone number by last name, first name, company name or some other means. Instant access to up-to-date information.

Scenario 5: Scheduling dinner with friends
You pick the time, the place and the friends. Unlike the preceding examples, dinner with friends may not be related to business. It is unlikely that all your friends share a common corporate calendar system.

Sync with personal info	Real-time group info
Call all the friends, play phone or email tag until the dinner is arranged. Everyone places the event separately on their calendars.	Create an invite and send a message to your friends to accept from their phones or email. They don't even have to be on the same system.

At the 11th hour, you find out Mary is a recently converted vegetarian and the steak place you picked might be offensive.

Sync with personal info	Real-time group info
Call all the friends, play phone or email tag until the change in the dinner location is arranged.	Change the meeting on your calendar and all invitees will be notified by message, email or voice.

Scenario 6: Planning a surprise vacation for your spouse

You want a surprise vacation. You either have access to your spouse's calendar or you ask for open times.

Sync with personal info	Real-time group info
There is no way to share information. Either sneak a peak at his/her organizer or pretend you have dinner plans at the in-laws.	Since your spouse has granted you access to his/her calendar, find an open weekend, schedule a weekend at the in-laws and a group system will request a placement on his/her calendar.

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Real-time, All The Time. The Way to Manage Your Information

NotWired believes that tomorrow's main conduit to the Internet will be through a mobile device using voice and all browser type interfaces. You will interact with this device to communicate, collaborate, schedule, make notes to yourself, and do things we are just beginning to imagine. We work and socialize as a group. We share information. Any system that limits sharing of information is archaic.

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